sciendo

PU.M.A. Vol. **30** (2022), No. 3, pp. 54–64, DOI:10.2478/puma-2022-0024

On the construction of the elementary trigonometric functions

Péter Medvegyev*
Department of Mathematics
Corvinus University of Budapest
Budapest, Hungary

(Received: July 22, 2022)

Abstract

The article discusses the construction of the elementary trigonometric functions. It discusses several approaches, but the main message is that to construct the trigonometric functions one needs to follow the same approach as one should use during the construction of the real exponential function, but one should use complex numbers. The key point is that one must construct the complex square root function to define the trigonometric functions on the binary numbers and then one should use some convexity argument to prove their differentiability. This approach, as the power series approach, based on the intimate relation between the trigonometric and the exponential functions.

Mathematics Subject Classifications (2015). 15A15

Keywords. elementary trigonometric functions

1 Formulation of the problem

The evergreen problem of an introductory course in calculus is the construction of the so called elementary functions. The construction of the real exponential function a^x is relatively easy. It is also not difficult to show that the so called Cauchy identity

$$a^{x+y} = a^x a^y \tag{1}$$

holds. It is also easy to show that the function is continuous. The first problem one faces is that one needs to prove that a^x is differentiable and one should show the formula

$$\frac{d}{dx} a^x = \ln a \cdot a^x.$$

A much bigger problem is that one should also introduce the trigonometric functions $\sin x$ and $\cos x$ and one should show that these functions are also differentiable and one should calculate their derivatives. The main problem is

*e-mail: medvegyev@uni-corvinus.hu, peter.medvegyev@gmail.com

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that to use the definitions and properties introduced in the high school one must define the length of an arc for which one needs the concept of integration. It is a bit strange that one needs to wait more or less up to the very end of the course to give the definition of the most elementary functions. A popular workaround is that one defines the power series of the complex exponential function and introduces the trigonometric functions as the value of the complex exponential function along the imaginary axis. [1], [2] The main problem with this approach is that it is a bit too formal, one can even say that too elegant and it is conceptually not really clear what is the relation of the circle and the trigonometric functions. Another problem is that one needs to know quite a lot about power series, which is also an advanced topic of the elementary analysis, perhaps it is even more advanced than integral calculus. In this note using the ideas of Bourbaki [3] I try to show a unified approach and I try to show that the construction of the real exponential function and the trigonometric functions are based on very similar ideas.

2 Constructing the functions: taking square roots iteratively

The construction of a^x is very simple: Let $a > 0$ be an arbitrary real number. By definition let $a^0 = 1$ and $a^1 = a$. Also by definition let $a^{1/2} = \sqrt{a}$ and taking square roots successively let $a^{1/2^{n+1}} = \sqrt{a^{1/2^n}}$. Then using (1) one can extend the definition for every binary number $k/2^n$. Of course k can be any negative or positive integer. It is obvious that the definition is correct that is if $k/2^m = 1/2^n$, then they provide the same number. From the construction it is obvious that a^x is increasing/decreasing, hence one can take the limit from the left and this way one can extend a^x for every real number x. It is also clear that identity (1) remains valid after the extension. As

$$a^{x+h} - a^x = a^x \left(a^h - 1\right),$$

to prove the continuity of a^x it is sufficient to show that a^x is continuous at $x = 0$. As the function is monotonous to calculate the limit it is sufficient to take any well chosen sequence. As $\sqrt[n]{a} \to 1$

$$\begin{aligned}
\lim_{h \searrow 0} a^h &= \lim_{n \to \infty} a^{1/2^n} = 1 = a^0, \\
\lim_{h \nearrow 0} a^h &= \lim_{h \nearrow 0} \frac{1}{a^{-h}} = \frac{1}{\lim_{h \searrow 0} a^h} = a^0.
\end{aligned}$$

Observe that to prove continuity it is sufficient to prove that a^x continuous from the right, the continuity from the left already follows from (1). If $a \neq 1$, then a^x not only monotonous but strictly monotonous hence it has a continuous inverse called $\log_a x$. One should observe that a^x and $\log_a x$ satisfy all the relations one

has learnt in the high school:

$$\begin{aligned}\log(ab) &= \log a + \log b, \quad \log a^b = b\log a,\\ a^{x+y} &= a^x a^y, \quad (a^x)^y = a^{xy},\\ (ab)^x &= a^x b^x, \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}.\end{aligned}$$

Our goal is to introduce the trigonometric functions in an entirely analogous way. The only difference is that we will use the unit circle of the complex plain. From the point of view of elementary geometry starting from the right angle one can naturally calculate certain values of the trigonometric functions with successive angular bisection. As a first step we want to construct a function $E(x) = \cos_1 x + i \sin_1 x$ which satisfies the Cauchy identity and

$$E(0) = 1 = (1,0), \quad E(1/4)) = i = (0,1).$$

By the Cauchy identity

$$E(0) = 1, \quad E(1/4) = i, \quad E(1/2) = -1, \quad E(1) = 1,$$

therefore $E(x)$ will be periodical with period 1. As in the real case let

$$E\left(\frac{1}{2^{n+1}}\right) = \sqrt{E\left(\frac{1}{2^n}\right)}, \quad E\left(\frac{k}{2^n}\right) = \left(E\left(\frac{1}{2^n}\right)\right)^k.$$

On the complex plain one can easily calculate the square root. If $c, d \geq 0$, then by definition

$$a + bi = \sqrt{c + di}$$

where

$$a = \sqrt{\frac{c + \sqrt{c^2 + d^2}}{2}}, \quad b = \sqrt{\frac{-c + \sqrt{c^2 + d^2}}{2}}.$$

If we are on the unit circle of the first quadrant then

$$a = \sqrt{\frac{c+1}{2}}, \quad b = \sqrt{\frac{1-c}{2}}.$$

$E(1/4) = i$, hence $\cos_1(1/4) = \operatorname{Re} E(1/4) = 0$ and as $E(0) = 1$ one has that $\cos_1 0 = \operatorname{Re} E(1/4) = 1$. With induction

$$\begin{aligned}\cos_1\left(\frac{1}{2^{n+1}}\right) &= \sqrt{\frac{\cos_1(1/2^n) + 1}{2}} \geq \sqrt{\frac{\cos_1(1/2^n) + \cos_1(1/2^n)}{2}} =\\ &= \sqrt{\cos_1(1/2^n)} \geq \cos_1\left(\frac{1}{2^n}\right).\end{aligned}$$

That is the sequence $a_n = \cos_1(1/2^n)$ is increasing, that is on the set of points $1/2^n$ $\cos_1 x$ is decreasing. Hence the sequence $b_n = \sin_1(1/2^n)$ is decreasing,

therefore $\sin_1 x$ is increasing on the set of points $1/2^n$. One can extend this to all dyadic points of $[0, 1/4]$. Really, with induction, using the multiplication rule of the complex numbers

$$\begin{aligned} 0 &\leq \cos_1 \frac{k+1}{2^n} = \operatorname{Re}\left(E\left(\frac{k+1}{2^n}\right)\right) = \operatorname{Re}\left(E\left(\frac{k}{2^n}\right)E\left(\frac{1}{2^n}\right)\right) = \\ &= \cos_1\left(\frac{k}{2^n}\right)\cos_1\left(\frac{1}{2^n}\right) - \sin_1\left(\frac{1}{2^n}\right)\sin_1\left(\frac{k}{2^n}\right) \leq \\ &\leq \cos_1\left(\frac{k-1}{2^n}\right)\cos_1\left(\frac{1}{2^n}\right) - \sin_1\left(\frac{1}{2^n}\right)\sin_1\left(\frac{k-1}{2^n}\right) = \\ &= \cos_1 \frac{k}{2^n}. \end{aligned}$$

$$\sin_1 \frac{k+1}{2^n} = \sqrt{1-\cos_1^2 \frac{k+1}{2^n}} \geq \sqrt{1-\cos_1^2 \frac{k}{2^n}} = \sin_1 \frac{k}{2^n}.$$

As in the exponential case we can extend $\cos_1 x$ and $\sin_1 x$ to all points of $[0, 1/4]$. Taking limit in the iteration $a_{n+1} = \sqrt{(a_n+1)/2}$

$$\lim_{n\to\infty} a_n = a = \sqrt{\frac{a+1}{2}},$$

hence

$$a^2 - \frac{1}{2}a - \frac{1}{2} = 0,$$

which has two roots $a = 1$, $a = -1/2$. Therefore $E(1/2^n) \to 1$. Using the monotonicity of the functions it is obvious that $\cos_1 x$ and $\sin_1 x$ are continuous from the right at $x = 0$. Using

$$E(-x) = \frac{1}{E(x)} = \frac{\cos_1 x - i \sin_1 x}{\sqrt{\cos_1^2(x) + \sin_1^2(x)}} = \cos_1 - i \sin_1 x$$

one can extend $E(x)$ to the interval $[-1/4, 1/4]$ and obviously it is continuous at $x = 0$. If $x \in [1/4, 1/2]$, then $E(x) = E(1/4)E(x - 1/4)$ etc. Hence $E(x)$ is defined and satisfies the Cauchy identity on the whole real line. Therefore it is continuous everywhere. The angle addition formulae follow from the rules of the multiplication of the complex numbers.

3 Consequences of differentiability

1. Assume that f satisfies the Cauchy identity $f(x+y) = f(x)f(y)$. As

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = f(x) \lim_{h\to 0} \frac{f(h) - 1}{h},$$

to prove the differentiability of f one needs to show that f is differentiable at $x = 0$ and the derivative is always of the form $f'(x) = f(x) f'(0)$. As

$$\lim_{h\to 0} \frac{f(h) - f(0)}{h} = \lim_{h\to 0} \frac{f(h) - 1}{h} = \lim_{h\to 0} \frac{1}{f(h)} \frac{1 - f(-h)}{h} = \lim_{h\to 0} \frac{1}{f(h)} \frac{f(-h) - 1}{-h}$$

to prove the differentiability it is sufficient to show that the derivative from the right or from the left exists.

2. Assume that we have shown the differentiability of a^x and denote $\ln a$ the derivative of the function at $x = 0$. From the product differentiation rule and from the identity $(ab)^x = a^x b^x$ one has that $\ln ab = \ln a + \ln b$. Also $\ln 1 = 0$ and if $a > 1$ then a^x increasing so $\ln a > 0$. If $0 < a < 1$ then $\ln a < 0$. If $a \neq 1$ and $a > 0$ then $\ln a \neq 0$ so $e = a^{1/\ln a}$ is well defined.

$$\frac{d}{dx} e^x = \frac{d}{dx} \left(a^{1/\ln a}\right)^x = \frac{\ln a}{\ln a} e^x = e^x,$$

hence $\ln e = 1$. As $\ln a/b = \ln a - \ln b$ it is obvious that $\ln a$ is strictly increasing, so there is just one number for which $\ln a = 1$.

$$\frac{d}{dx} \left(e^{\ln a}\right)^x = \ln a \cdot \left(e^{\ln a}\right)^x$$

so $e^{\ln a} = a$ hence $\ln a = \log_e a$. Using the inverse function differentiation rule

$$\frac{1}{x} = \frac{d \ln x}{dx}, \quad x > 0,$$

hence if $z > 0$ and $x = 1/z$, and $\exp(x) = e^x$ then

$$\begin{aligned} e^z &= e^{1/x} = \exp\left(\lim_{h\to 0} \frac{\ln(x+h) - \ln(x)}{h}\right) = \\ &= \lim_{h\to 0} \exp\left(\ln\left(1 + \frac{h}{x}\right)^{1/h}\right) = \\ &= \lim_{u\to\infty} \left(1 + \frac{z}{u}\right)^u. \end{aligned} \tag{2}$$

One can extend this to any z. If $z < 0$, then

$$\begin{aligned} \exp(z) &= \frac{1}{\exp(-z)} = \frac{1}{\lim_{u\to\infty} \left(1 - \frac{z}{u}\right)^u} = \lim_{u\to\infty} \frac{1}{\left(1 - \frac{z}{u}\right)^u} = \\ &= \lim_{u\to\infty} \left(\frac{1}{1 - \frac{z}{u}}\right)^u = \lim_{u\to\infty} \left(\frac{u - z + z}{u - z}\right)^u = \\ &= \lim_{u\to\infty} \left(1 + \frac{z}{u - z}\right)^u = \lim_{u\to\infty} \left(1 + \frac{z}{u - z}\right)^{u-z} = \\ &= \lim_{u\to\infty} \left(1 + \frac{z}{u}\right)^u. \end{aligned}$$

3. Assume we have proved the differentiability of $E(x)$. As $|E(x)| = 1$

$$1 = |E(x)|^2 = (\operatorname{Re}(E(x)))^2 + (\operatorname{Im}(E(x)))^2 = u^2(x) + v^2(x).$$

Differentiating $2u \cdot u' + 2v \cdot v' = 0$, hence $\langle E(x), E'(x) \rangle = 0$. As $E(0) = 1 = (1, 0)$, there is a real number α such that $E'(0) = (0, \alpha) = \alpha i$ and

$$E'(x) = E(x) E'(0) = \alpha i E(x).$$

$\sin_1(x)$ is increasing, so the derivative of $\sin_1 x$ is non negative. If we introduce the $\pi = \alpha/2$ then $\pi > 0$. If $E_a(x) = E(x/a)$, then $E_a(x)$ is periodical with period a.

$$E'_{2\pi}(x) = \left(E\left(\frac{x}{2\pi}\right)\right)' = \frac{i2\pi}{2\pi} E_{2\pi}(x) = i \cdot E_{2\pi}(x).$$

If by definition $\sin x = \operatorname{Im} E_{2\pi}(x)$ and $\cos x = \operatorname{Re} E_{2\pi}(x)$, then these functions are periodical with period 2π and

$$(\cos x + i \sin x)' = i \cdot (\cos x + i \sin x) = -\sin x + i \cos x,$$

therefore

$$\frac{d \sin x}{dx} = \cos x, \quad \frac{d \cos x}{dx} = -\sin x.$$

From the construction $\cos \pi/2 = \operatorname{Re} E(1/4) = 0$, and $\sin \pi = \operatorname{Im} E(1/2) = 0$.

$$\cos \frac{\pi}{4} = \sqrt{\frac{\cos \pi/2 + 1}{2}} = \sqrt{\frac{1}{2}}.$$

As

$$\cos nx = \operatorname{Re} (\cos x + i \sin x)^n,$$

one has that

$$\cos 3x = 4 \cos^3 x - 3 \cos x.$$

From this

$$0 = \cos \frac{\pi}{2} = 4 \cos^3 \frac{\pi}{6} - 3 \cos \frac{\pi}{6},$$

hence

$$\cos^2 \frac{\pi}{6} = \frac{3}{4},$$

and as $\pi/6 < \pi/2$, that is $\cos \pi/6 > 0$ therefore

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}, \quad \sin \frac{\pi}{6} = \frac{1}{2}.$$

Hence

$$\frac{1}{2} = \sin \frac{\pi}{6} \leq \frac{\pi}{6},$$

as $\sin 0 = 0$ and $(\sin x)' \leq 1 = x'$. So $\pi \geq 3$. With differentiation it is easy to show $\sin x/x$ decreasing on the interval $[0, \pi]$ and

$$\lim_{x \to 0} \frac{\sin x}{x} = \frac{d}{dx} \sin x|_{x=0} = 1,$$

so

$$2^n \sin \pi/2^n \nearrow \pi,$$

hence

$$2^n \sin \frac{\pi}{2^n} = 2^n \sqrt{1 - \cos^2\left(\frac{\pi}{2^n}\right)} \leq \pi.$$

If $n = 10$ then starting from $\cos \frac{\pi}{4} = \sqrt{2}/2$ with iteration one can calculate $\cos^2\left(\frac{\pi}{2^n}\right)$ and one gets the estimation $\pi \geq 3.141572940370199$. One can use a slightly better estimation using

$$\sin x = 2 \sin \frac{x}{2} \cos \frac{x}{2} = \ldots = 2^n \sin \frac{x}{2^n} \prod_{k=1}^{n} \cos \frac{x}{2^k}.$$

Taking limit

$$\sin x = x \prod_{k=1}^{\infty} \cos \frac{x}{2^k}.$$

Now if $x = \pi/2$, then

$$\pi = \frac{2}{\prod_{k=1}^{\infty} \cos \frac{\pi}{2^{k+1}}} \geq \prod_{k=1}^{n} \cos \frac{\pi}{2^{k+1}}.$$

As an example

$$\pi \geq \frac{2}{\frac{\sqrt{2}}{2}\sqrt{\frac{1+\sqrt{2}/2}{2}}} = \frac{4}{\sqrt{1+\sqrt{2}/2}} \geq 3.$$

On the other hand

$$\lim_{n\to\infty} \frac{2^n}{\pi} \sin \frac{\pi}{2^n} \frac{1}{\cos^2 \frac{\pi}{2^{n+1}}} = 1.$$

The function

$$\frac{\sin x}{x \cos^2 x/2} = \frac{\sin x}{x} \frac{2}{1+\cos x}$$

is increasing as

$$\begin{aligned} \left(\frac{\sin x}{x\,(1+\cos x)}\right)' &= \frac{\cos x \cdot (x\,(1+\cos x)) - \sin x \cdot (1+\cos x - x \sin x)}{(x\,(1+\cos x))^2} = \\ &= \frac{(1+\cos x)\,(x - \sin x)}{(x\,(1+\cos x))^2} \geq 0. \end{aligned}$$

Hence

$$2^n \sin \frac{\pi}{2^n} \frac{1}{\cos^2 \frac{\pi}{2^{n+1}}} \geq \pi.$$

If $n = 1$, then

$$2 \sin \frac{\pi}{2} \frac{1}{\cos^2 \frac{\pi}{4}} = 4 \geq \pi.$$

If $n = 10$, then $\pi \leq 3.141593269696855$.

4 Proof of differentiability with convexity

To prove differentiability perhaps the simplest way is to use Jensen's theorem: if a continuous function is midpoint convex then it is convex. Recall that a convex function has left and right derivatives, which as a result of the Cauchy identity are equal which implies differentiability.

a^x is midpoint convex as using the inequality between arithmetic and geometric means

$$a^{\frac{x+y}{2}} = \sqrt{a^x a^y} \leq \frac{1}{2}\left(a^x + a^y\right).$$

To show the differentiability of $E(x)$ it is sufficient to show that $\cos_1 x$ and $\sin_1 x$ are midpoint concave on the interval $[0, 1/4]$. As on this interval $\cos_1 x \geq 0$

$$\cos_1\left(\frac{x+y}{2}\right) = \operatorname{Re} E\left(\frac{x+y}{2}\right) =$$

$$\operatorname{Re}\sqrt{E(x+y)} = \sqrt{\frac{\operatorname{Re} E(x+y)+1}{2}} = \sqrt{\frac{\cos_1(x+y)+1}{2}} =$$

$$\sqrt{\frac{\cos_1 x \cos_1 y - \sin_1 x \sin_1 y + 1}{2}} = \sqrt{\frac{\left(\cos_1 x + \cos_1 y\right)^2 + \left(\sin_1 x - \sin_1 y\right)^2}{4}} \geq$$

$$\sqrt{\frac{\left(\cos_1 x + \cos_1 y\right)^2}{4}} = \frac{1}{2}\left(\cos_1 x + \cos_1 y\right).$$

The same is working for $\sin_1 x$

$$\sin_1\left(\frac{x+y}{2}\right) = \operatorname{Im} E\left(\frac{x+y}{2}\right) =$$

$$\operatorname{Im}\sqrt{E(x+y)} = \sqrt{\frac{1-\operatorname{Re} E(x+y)}{2}} = \sqrt{\frac{1-\cos_1(x+y)}{2}} =$$

$$\sqrt{\frac{1-\cos_1 y \cos_1 x + \sin_1 x \sin_1 y}{2}} = \frac{1}{2}\sqrt{\left(\sin_1 x + \sin_1 y\right)^2 + \left(\cos_1 x - \cos_1 y\right)^2} \geq$$

$$\frac{1}{2}\left(\sin_1 x + \sin_1 y\right).$$

5 Using integration

Using elementary properties of integration Bourbaki shows that under very general conditions if a continuous function satisfies Cauchy identity then the function is automatically differentiable. Although this is a nice observation as the proof is based on integration it is not of a great help. Using integral calculus one can very easily and very naturally introduce the elementary functions. [4]

1. In this approach using integral calculus we first define the inverse functions. By definition

$$\ln x = \int_1^x \frac{1}{t} dt.$$

so

$$\frac{d \ln}{dx} = \frac{1}{x}.$$

By the compose function differentiation rule

$$\frac{d}{dx} \ln ax = \frac{1}{ax} a = \frac{1}{x}.$$

As the derivative of $\ln ax$ and $\ln x$ are the same

$$\ln ax = \ln x + c.$$

If $x = 1$, then $\ln a = \ln 1 + c = c$, so

$$\ln ab = \ln a + \ln b.$$

$\ln x$ is continuous and it is strictly increasing so it has an inverse $\exp(x)$. Obviously

$$\begin{aligned} \exp(x+y) &= \exp(x) \exp(y), \\ \frac{d \exp(x)}{dx} &= \exp(x). \end{aligned}$$

2. As we have the tool of integration we can introduce the length of an arc. By definition the length of the arc defined by a function $f(x)$ on $[a, b]$ is

$$\int_a^b \sqrt{1 + (f'(x))^2} dx.$$

Using this for $f(x) = \sqrt{1 - x^2}$

$$\int_a^b \sqrt{1 + \left(\frac{-2x}{2\sqrt{1-x^2}} \right)^2} dx = \int_a^b \sqrt{1 + \frac{x^2}{1-x^2}} dx = \int_a^b \frac{1}{\sqrt{1-x^2}} dx.$$

If $a = 0$ and $-1 < b < 1$ then this formula as function of b defines a strictly increasing continuous function called arcsin. A minor problem is that we want to extend it to the interval. $[-1, 1]$. The integral

$$\int_0^x \frac{1}{1+t^2} dt, \quad -\infty < x < \infty,$$

is convergent. With substitution

$$t = \frac{u}{\sqrt{1-u^2}}, \quad -1 < u < 1$$

one has with direct calculation

$$\int_0^{x/\sqrt{1+x^2}} \frac{1}{1+\frac{u^2}{1-u^2}} \cdot \frac{\sqrt{1-u^2}-u\frac{-2u}{2\sqrt{1-u^2}}}{1-u^2} du = \int_0^{x/\sqrt{1+x^2}} \frac{1}{\sqrt{1-u^2}} du.$$

So

$$\int_0^x \frac{1}{1+t^2} dt = \arcsin \frac{x}{\sqrt{1+x^2}}.$$

If $x \to \pm\infty$ then arcsin can be extended as a finite function to ± 1. By definition let $\pm\pi/2$ be the extended values. Let $\sin x$ be the inverse of $\arcsin x$. Obviously $\sin \pm\pi/2 = \pm 1$. Using the inverse function differentiation rule by definition

$$\frac{d \sin x}{dx} = \sqrt{1-\sin^2 x} = \cos x, \quad -\pi/2 < x < \pi/2.$$

Using the differentiation rule for the composed functions if $-\pi/2 < x < \pi/2$

$$\frac{d \cos x}{dx} = -\frac{1}{2\sqrt{1-\sin^2 x}} 2 \sin x \frac{d \sin x}{dx} = -\sin x.$$

The derivatives at the end points, using the continuity of $\cos x$ and the mean value theorem

$$\lim_{x \nearrow \pi/2} \frac{\sin x - \sin \pi/2}{x - \pi/2} = \lim_{x \nearrow \pi/2} \cos \xi = \cos \frac{\pi}{2} = 0.$$

The calculation of the other derivatives is similar. Let

$$\begin{aligned} f(x) &= (\sin(a+x) - \sin a \cos x - \cos a \sin x)^2 + \\ &\quad + (\cos(a+x) - \cos a \cos x + \sin a \sin x)^2 = \\ &= A^2 + B^2. \end{aligned}$$

As $\cos 0 = 1$, $\sin 0 = 1$ $f(0) = 0$.

$$\frac{df(x)}{dx} = \frac{d}{dx}\left(A^2 + B^2\right) = 2AA' + 2BB',$$

where

$$\begin{aligned} A' &= \cos(a+x) + \sin a \sin x - \cos a \cos x = B \\ B' &= -\sin(a+x) + \cos a \sin x + \sin a \cos x = -A, \end{aligned}$$

hence $\frac{df}{dx} = 0$. Therefore $f \equiv 0$, which proves the angle addition formulae. Using these identities one can extend the trigonometric functions to the whole real line.

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